

WASP-10b

R-1951

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_130617 · created 2026-08-06T15:37:46+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456460.9159 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.211 +/- 0.061
Transit depth (Rp/Rs)^2	4.47 +/- 2.58 [%]
Orbital Inclination (inc)	89.98 +/- 1.89
Airmass coefficient 1 (a1)	4.02 +/- 3.61
Airmass coefficient 2 (a2)	-0.56 +/- 0.65
Scatter in the residuals of the lightcurve fit is	106.21 %
Best Comparison Star	#3 - [218, 244]
Optimal Aperture	0.75
Optimal Annulus	5.0
Transit Duration (day)	0.0945 +/- 0.0099

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.211 ± 0.061 vs 0.1592 (deviation 0.85σ)
Duration fit vs published	0.0945 d vs 0.0946 d (deviation 0.01σ)
Mid-time O–C	3.9 min
Depth SNR	3.5
Residual scatter	106.21 %

Light curve

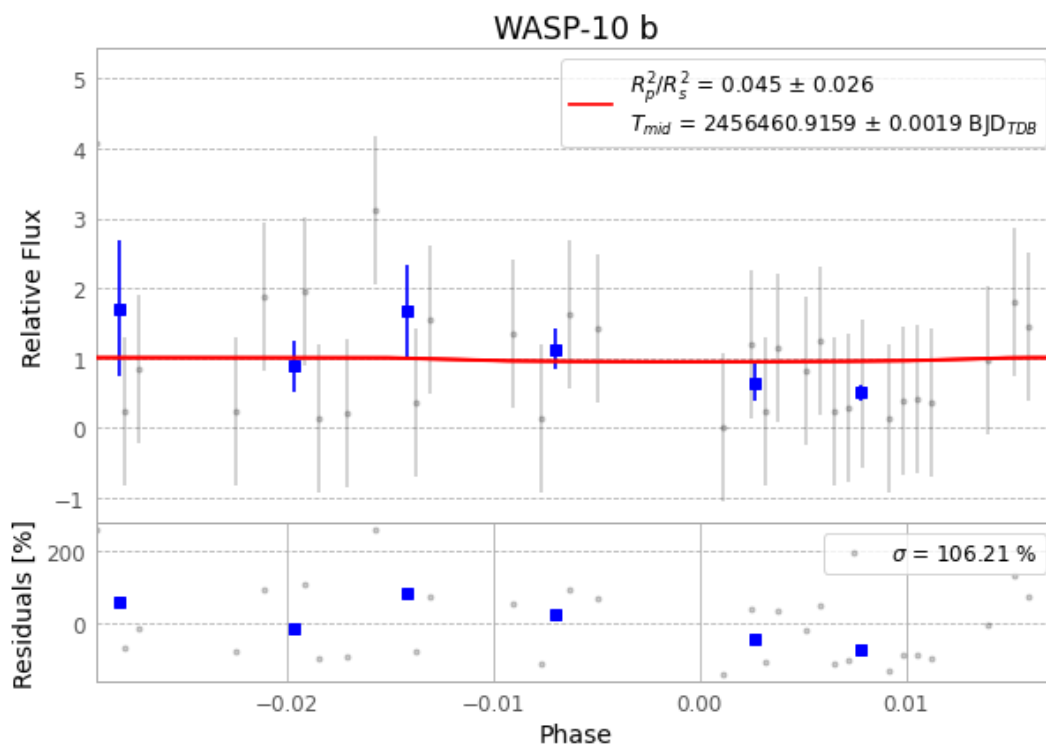


Figure 1 — FinalLightCurve_WASP-10 b_2013-06-17.png (SHA-256 5aac2e671c7b25e4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [495, 215], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 db7f46fea37bbc3b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

71 FITS frames (47 MB), WASP-10130617074512.FITS → WASP-10130617111512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-130617.log (b5bc4eda06f0...), FinalLightCurve_WASP-10 b_2013-06-17.png (5aac2e671c7b...), FinalLightCurve_WASP-10 b_2013-06-17.csv (6fa5c3e910f3...)

Compute resources

Wall time 112.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 15:37Z.